

Working Memory Involvement in Higher Cognition: Insights from fMRI Modeling

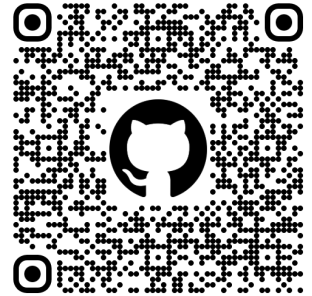
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Zeb Caslick-Waller, Wendi Li, Tony Gao, Raúl Rodríguez Cruces



Montbretia Cabinet



Meet the team!



Meet the Team!

Member	Where could you find them?
Baitong Mu	ORCID , GitHub
Carmen Tang	ORCID , GitHub
Zeb Caslick-Waller	GitHub
Wendi Li	ORCID
Tony Gao	ORCID
Raúl Rodríguez Cruces	ORCID , GitHub
Saameh Sanaaee	ORCID , LinkedIn , Bluesky



Project Overview (Saameh)

Working Memory Architecture and Demand Model

This project is the result of work from the **Montbretia Cabinet** team during the Neuromatch Academy [Computational Neuroscience Course](#) and the Neuromatch Academy [Impact Scholars Program](#).

For information on using this repository, please refer to [this section](#).

Project Summary

Working memory (WM) allows us to temporarily hold information “online”, supporting higher cognitive functions such as decision-making, attention, and problem-solving. A key brain region responsible for WM is the prefrontal cortex, an area that differentiates us from other animals. Although our WM resources are limited, it adapts dynamically to the specific demands of different tasks.

Current study uses task-based fMRI data from a large-scale project, the Human Connectome Project, to investigate WM across a range of tasks and conditions. Task-based fMRI measures brain activation during WM tasks, where subjects are asked to recall images presented several trials before.

In our project, statistical and deep learning models acted as “sensors” for task demand and WM activity, predicting how much WM is involved in emotion and language domains. This work provides insights into WM theories from both computational and neurocognitive perspective, with significant implications for education and cognitive rehabilitation.



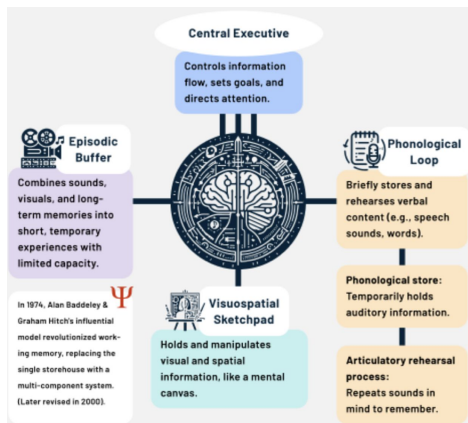
https://github.com/saamehsanaae/WMAD-Montbretia_Cabinet-ISP

Background - WM (Barbara)

Working Memory

- A cognitive system to temporarily hold and manipulate information (Baddeley, 2010)
- Supports human higher cognitive processes, such as reasoning, learning, problem solving, decision-making, language comprehension, etc.

Multicomponent Model



(Baddeley and Hitch, 1974, 2020)

Dynamic Neural Resources Model

- Conceptualize WM as emerging from the **dynamic allocation** of neural resources across a **distributed network**
- WM processes are **not confined** to fixed components but **adaptive** and **context-dependent**

(Tomić & Bays, 2024)

Which One
Explains
Better?

Background - WM: Emotion & Language (Barbara)

Research has suggested the **interactions** between:

- WM and Emotion in the prefrontal and subcortical regions, e.g., amygdala (Grimm et al., 2012; Lopez-Franco et al., 2018).
- WM and Language in subcortical structures and widespread right hemispheric regions (Deldar et al., 2020)

→ Our preliminary model (MLP) has demonstrated the different levels of WM involvement in language tasks (story vs. math).

However! The MLP did not capture temporary and relational patterns in fMRI data

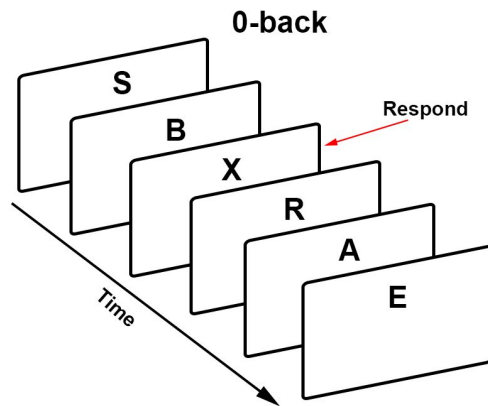
IMPROVE IT

Current Study (Barbara)

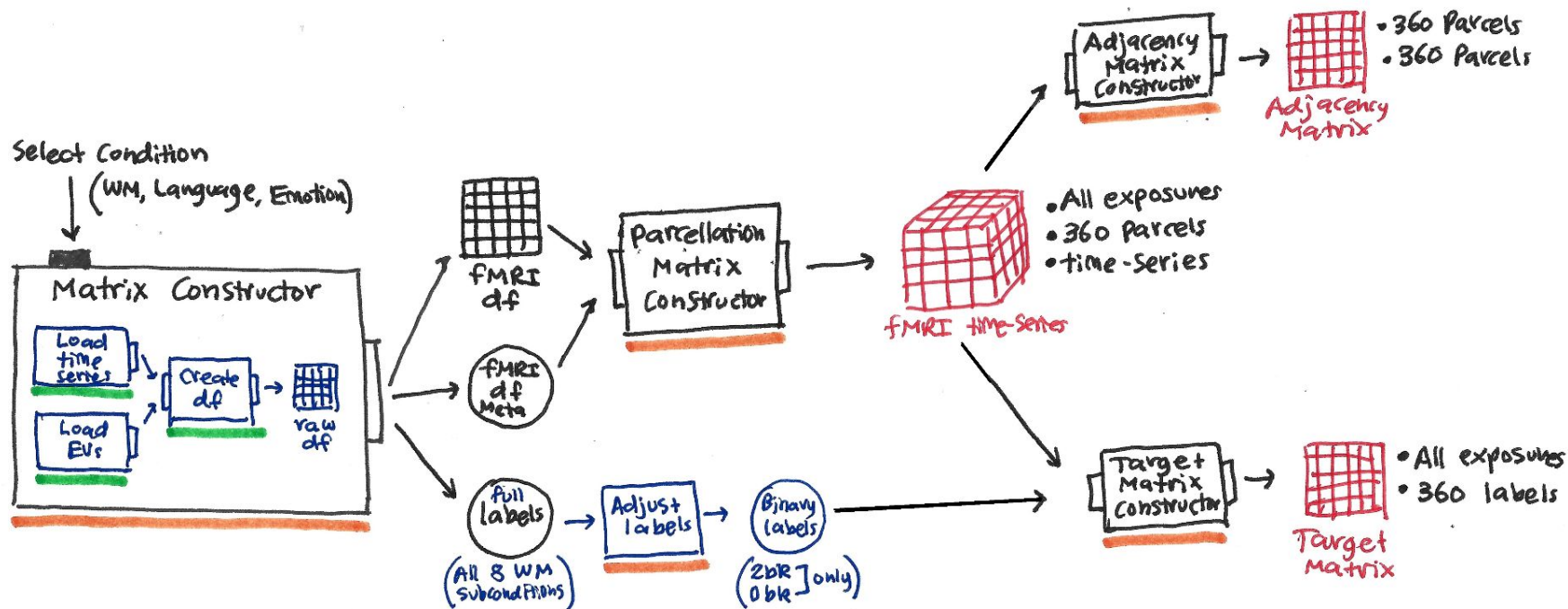
- Finding activation pattern of WM – Generalized Linear Model (**GLM**)
- Improving preliminary model by incorporating – Graph Neural Network (**GNN**) and Long Short-Term Memory (**LSTM**)

Main Dataset: **Human Connectome Project (HCP)**

- 100 healthy young adults
- Target Task (Subtask)
 - Working Memory (Body, Faces, Places, Tools)
 - Emotion (Neutral, Fear)
 - Language (Story, Math)
- Conditions
 - 0-back and 2-back demands
- 360 Parcels (Glasser), BOLD signals → **Parcel-based BOLD signals**

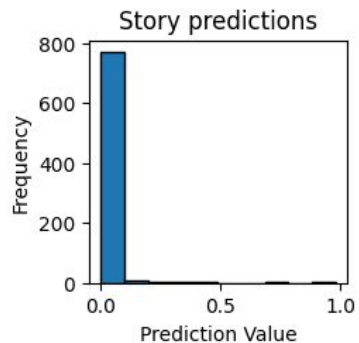
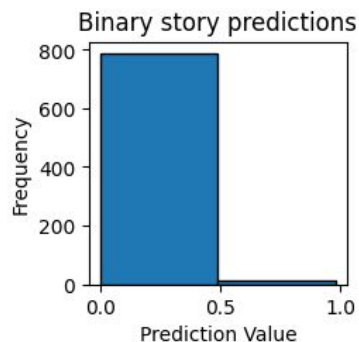
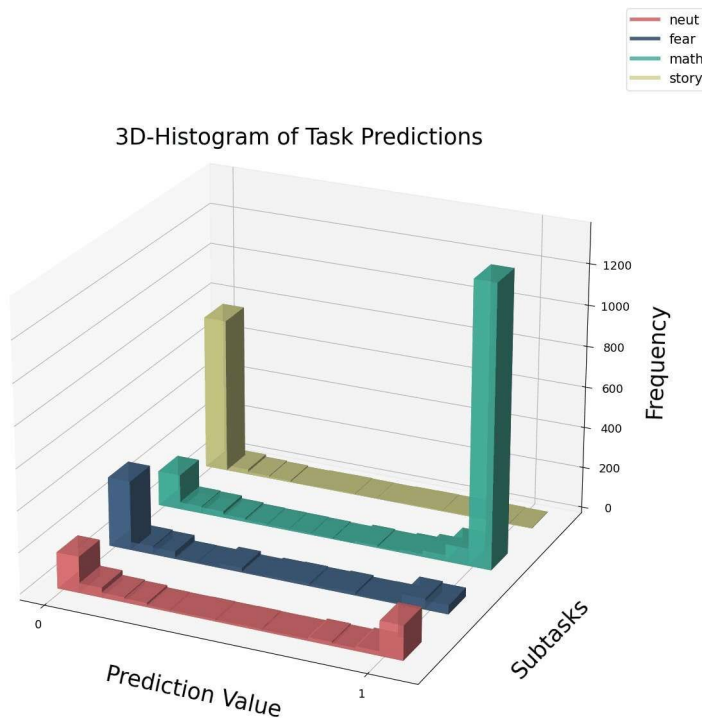
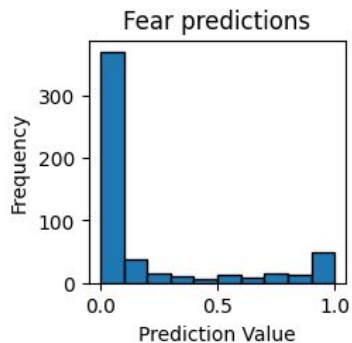
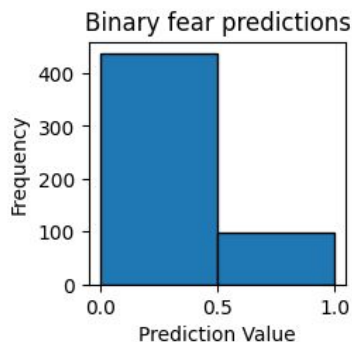


Methods - data and visualization (Zeb)

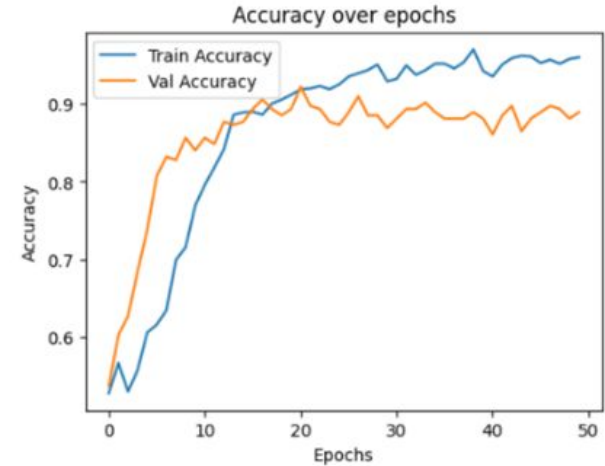
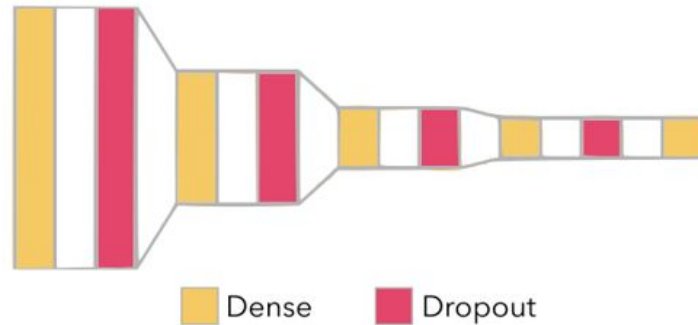
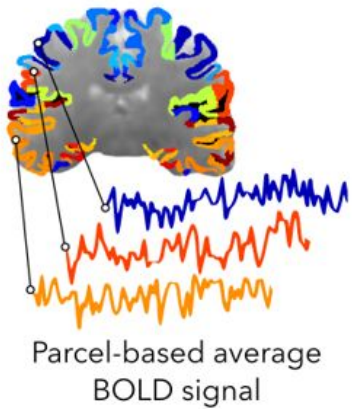


WMD Model (Saameh)

The Preliminary Model



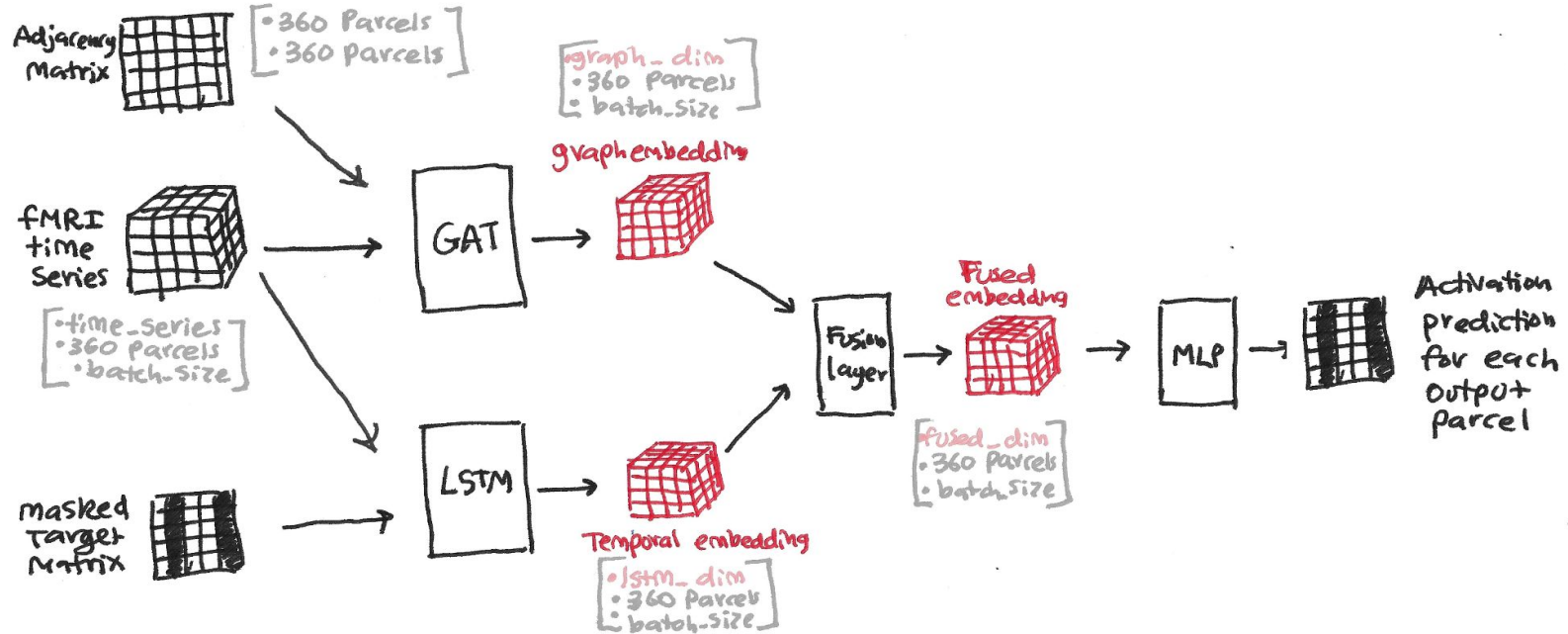
WMD Model (Saameh)

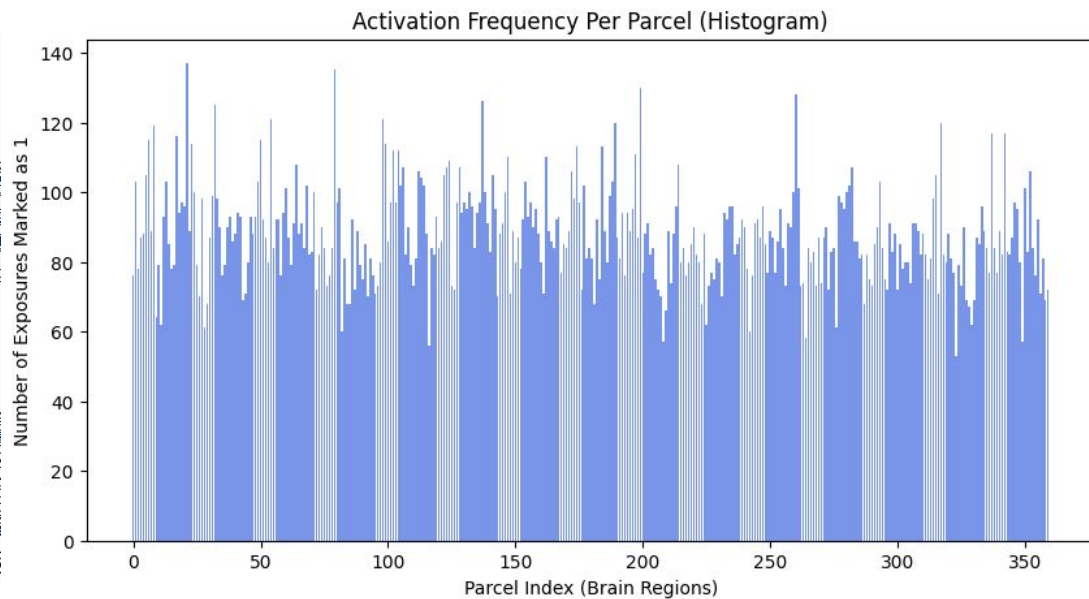
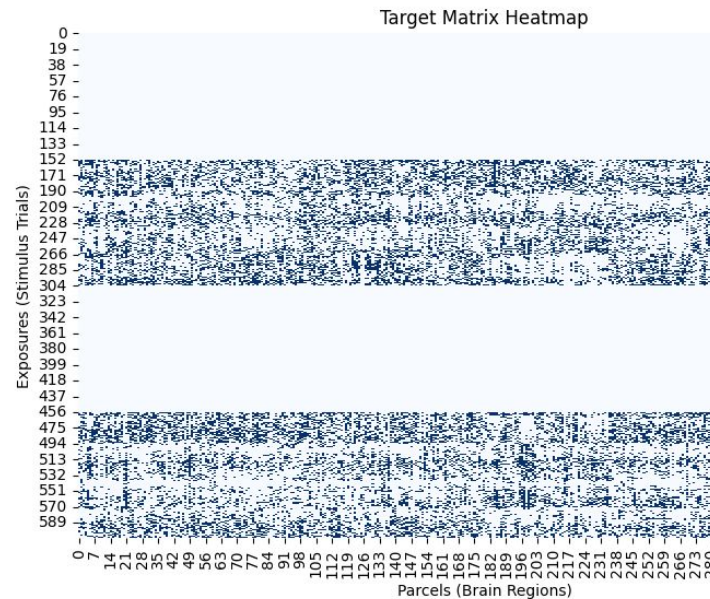


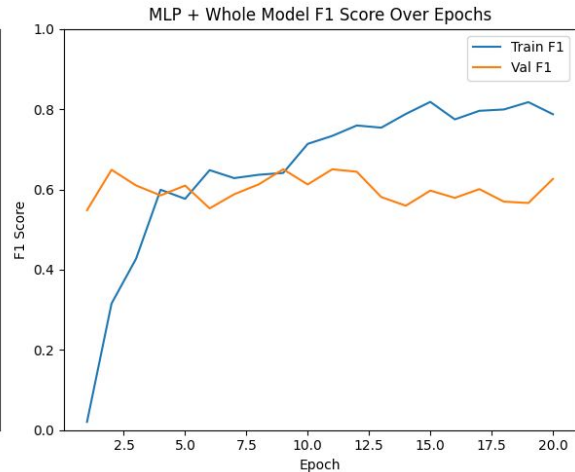
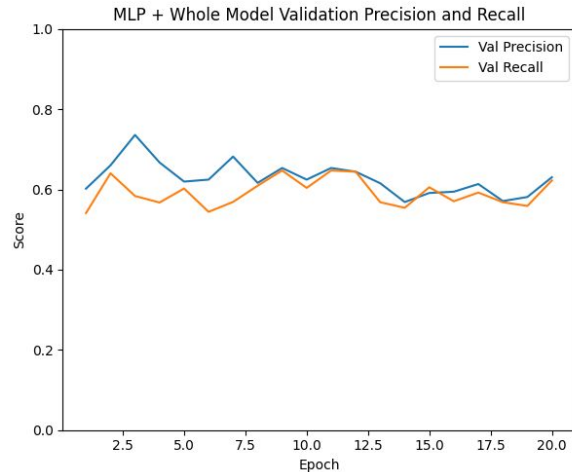
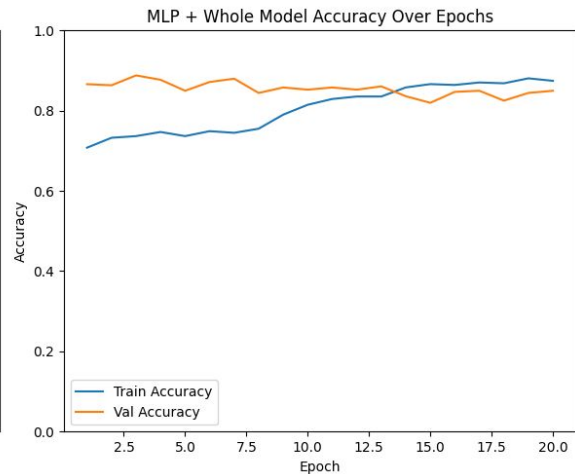
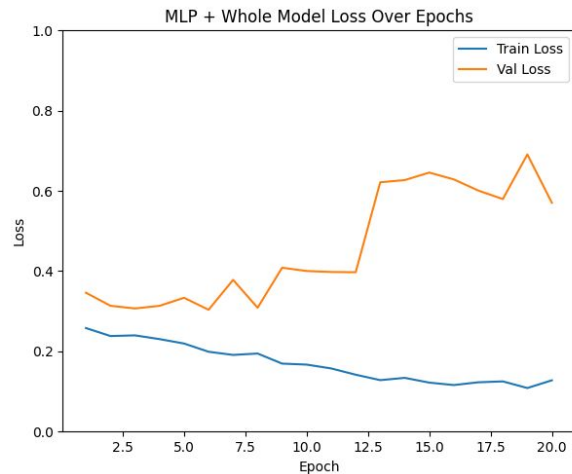
- Structure of the model
 - Funnel-shape for feature extraction
 - Dense and dropout layers

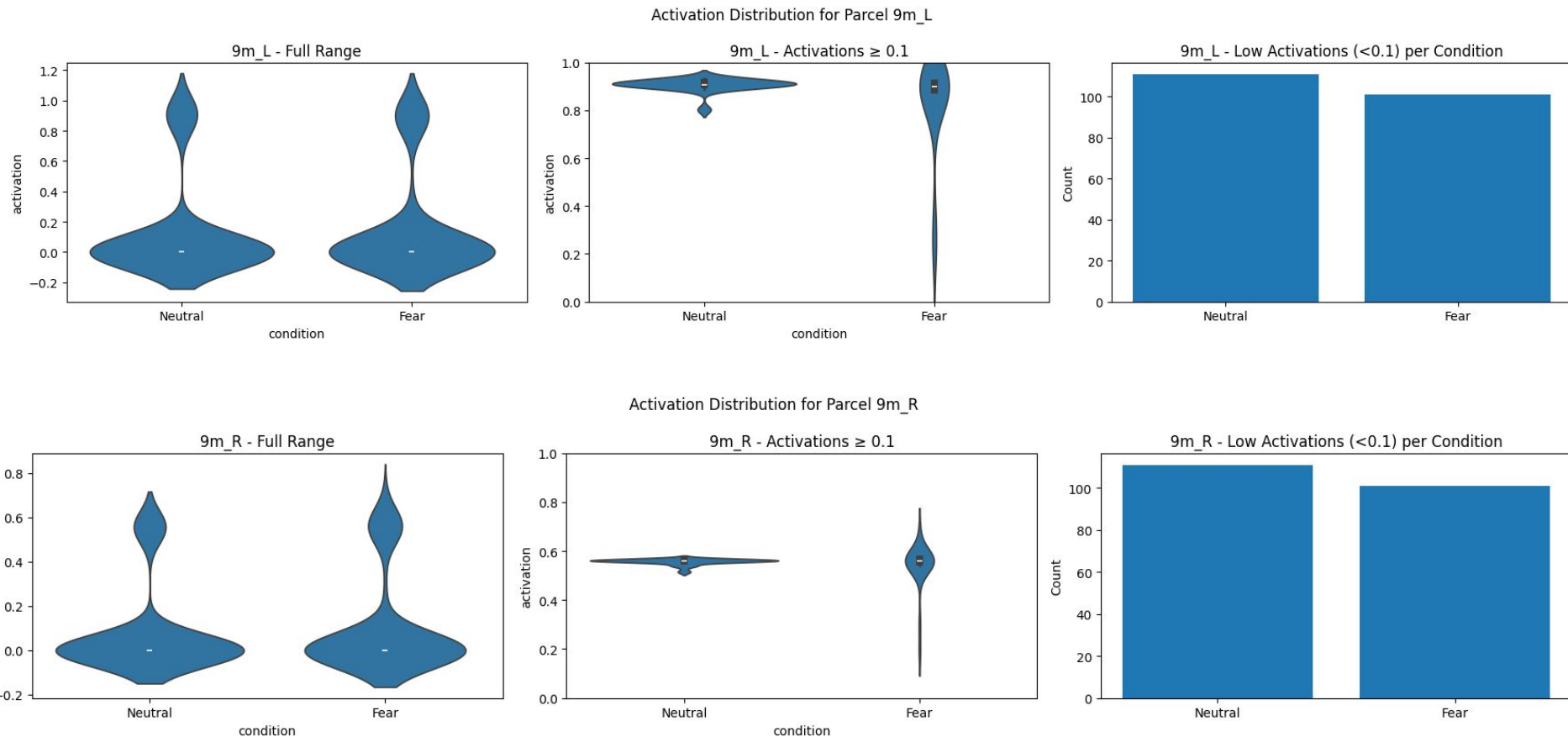
- Limitations & Reasons for improvement
 - Data, of course ...
 - Average BOLD signal
 - Paired output
 - Model architecture
 - Interpretability

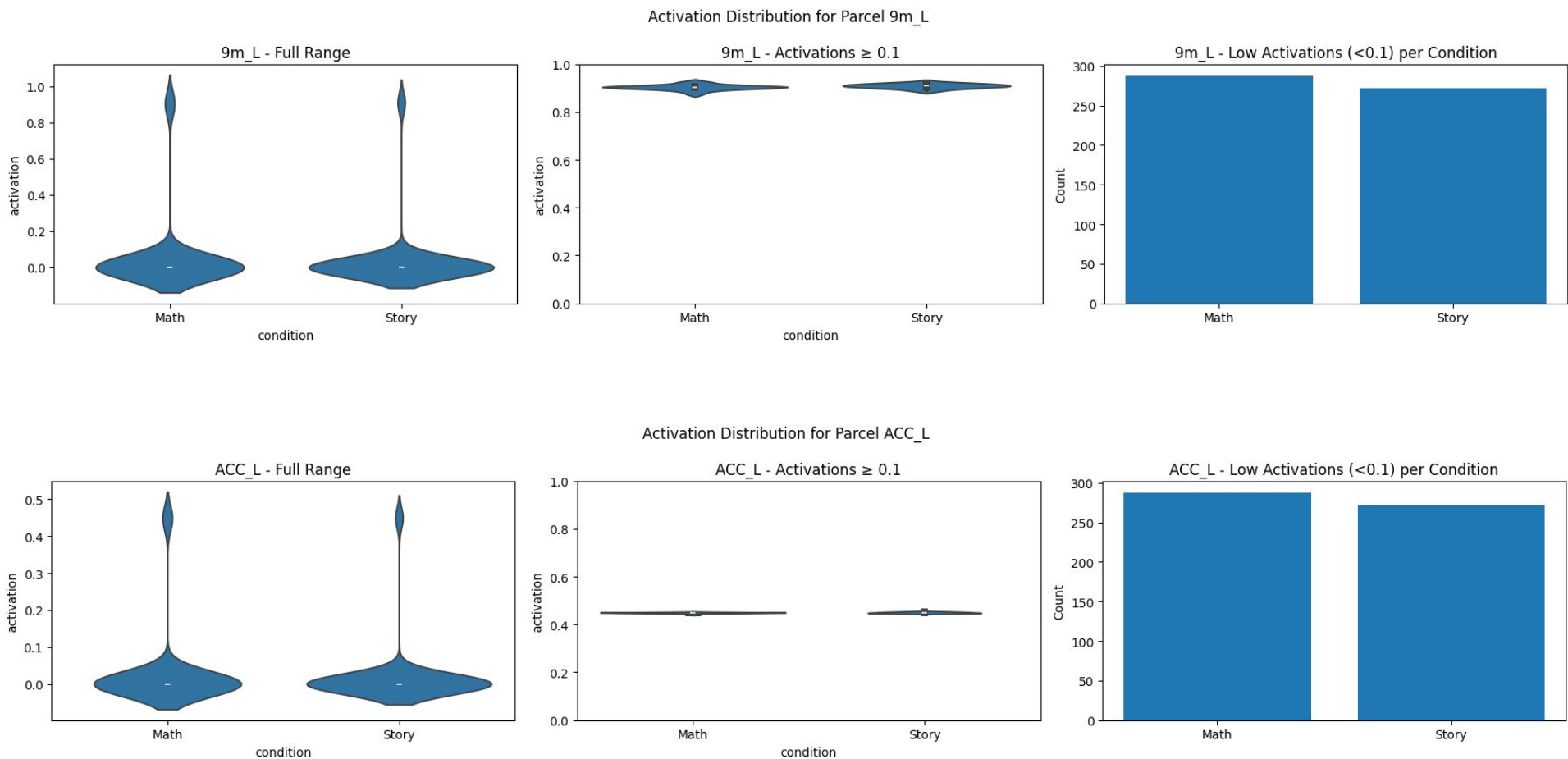
WMAD Model - GNN-LSTM (Zeb)



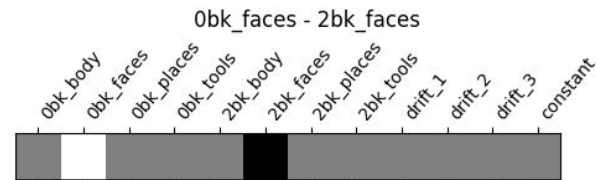
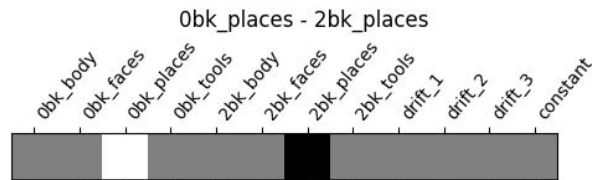
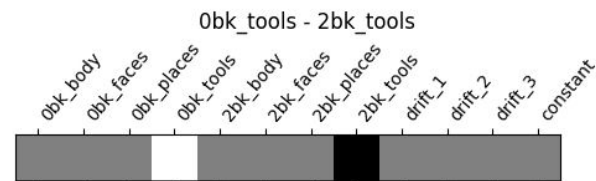
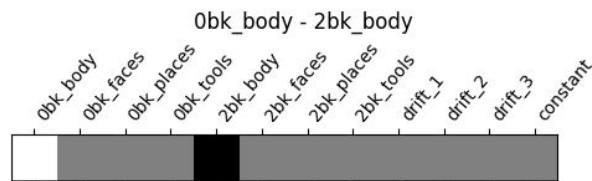
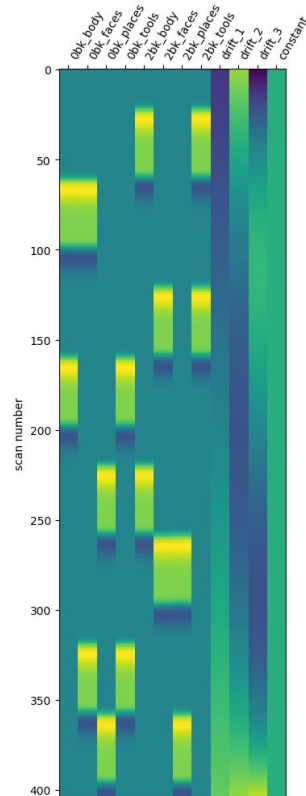








WMAD - GLM (Barbara)



GLM fitting

- 2D Event-based design matrix
- 2D parcellated BOLD time series array

Contrasts: 0-back vs. 2-back conditions

Discussion (Saameh)

- What did we find?
- What could this mean?
- Limitations!
- Future directions ...
 - Input
 - Model Architecture
 - Validation
 - Behavioural data
 - Population

“So what?”

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Thank you!

Commence the Q&A ...

